

Roll No.

TIT-704

B. TECH. (CS/IT)

(SEVENTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2022

CRYPTOGRAPHY AND NETWORK SECURITY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) List and briefly define the types of cryptanalytic attacks based on what is known to the attacker ? Also define the security goals.

6 + 4 Marks (CO1)

OR

- (b) (i) Classify the attacks in the following scenarios. Also justify your answer :

(A) Eva stole the bank card and PIN of Alice and pretended that she is the Alice.

(B) Alice has transferred to ₹ 1,000 to Bob and later founds out Bob has received only ₹ 500.

5 Marks

- (ii) Differentiate between block cipher and stream cipher.

5 Marks (CO1)

P. T. O.

(2)

TIT-704

2. (a) Use Playfair cipher to encrypt the message "THISISEASY", with the key "CRYPTO". 10 Marks (CO1)

OR

- (b) Encrypt the given plaintext by using the hill cipher. 10 Marks (CO1)

Plaintext : PAYMOR

Key:

17	17	5
21	18	21
2	2	19

3. (a) (i) Illustrate why the modern block ciphers are designed as substitution cipher instead of transposition ciphers. 5 Marks (CO2)
- (ii) Describe session key, master key and nonce. 5 Marks (CO3)

OR

- (b) (i) In the context of statistical cryptanalysis, what do you understand by the terms diffusion and confusion? 5 Marks (CO2)
- (ii) What type of information an opponent can derive from traffic analysis? Also explain the use of traffic padding. 3 + 2 Marks (CO3)

4. (a) Illustrate various block cipher modes of operation. 10 Marks (CO2)

OR

- (b) Describe how the following steps are used in AES : add key, substitution, shift row, and mix column. 10 Marks (CO2)

(3)

5. (a) Analyse the ways in which secret key can be distributed to two communicating parties.
10 Marks (CO3)

OR

- (b) Explain the two criteria used to validate the number is random. Also argue the importance of selecting the values a , c and m in linear congruential generator.
3 + 7 Marks (CO3)